

You are a Tier-1 SOC Analyst. Analyze the following security events. Event 1 Windows Event ID 4625 Multiple failed login attempts from IP 192.168.10.25 against user Administrator. Event 2 Windows Event ID 4624 Successful login for Administrator from the same IP five minutes later. Event 3 Sysmon Event ID 1 powershell.exe executed with an encoded command. Event 4 Sysmon Event ID 3 powershell.exe established an outbound HTTPS connection to IP 203.0.113.50 on port 443. For each event provide: • Explanation • Suspicious indicators • Indicators of Compromise (IoCs) • Severity • Recommended investigation steps Finish with an Overall Assessment summarizing the findings.

## Tier-1 SOC Investigation Report

### Incident Summary

The sequence of events suggests a **potential successful compromise** of the **Administrator** account followed by suspicious PowerShell activity and outbound network communication. The timeline is consistent with an attacker gaining access through brute force or password guessing, executing an encoded PowerShell payload, and communicating with an external system.

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### Event 1

#### Windows Event ID 4625

**Multiple failed login attempts from IP:** 192.168.10.25

**Target Account:** Administrator

#### Explanation

Windows Security Event ID **4625** indicates a failed authentication attempt. Multiple failed logins against the **Administrator** account may indicate:

- Incorrect password attempts
- Password spraying
- Brute-force attack
- Unauthorized access attempts

Repeated failures are far more suspicious than a single failed login.

#### Suspicious Indicators

- Multiple authentication failures
- Targeting the privileged Administrator account
- Same source IP for repeated attempts
- Potential password guessing activity

#### Indicators of Compromise (IoCs)

- Source IP: 192.168.10.25
- Target Account: Administrator
- Multiple failed authentication events

- Repeated login failures within a short time

### Severity

### Medium

*(High if numerous attempts occur in a short period or originate from an unexpected host.)*

### Recommended Investigation Steps

1. Determine the number of failed login attempts.
  2. Identify whether the source IP belongs to an internal workstation or unauthorized device.
  3. Review account lockout events.
  4. Check for similar attempts against other privileged accounts.
  5. Verify whether the Administrator account should be used from this system.
  6. Review authentication logs before and after the failures.
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### Event 2

#### Windows Event ID 4624

#### Successful Administrator login

Source IP: 192.168.10.25

#### Explanation

Windows Event ID **4624** records a successful authentication.

Since the successful login occurred **five minutes after multiple failed attempts from the same IP**, this is highly suspicious and may indicate:

- Successful brute-force attack
- Correct password eventually guessed
- Stolen credentials
- Legitimate user after mistyping password (less likely when combined with later events)

#### Suspicious Indicators

- Successful login immediately after repeated failures
- Privileged Administrator account
- Same originating IP
- Authentication sequence consistent with credential compromise

#### Indicators of Compromise (IoCs)

- Source IP: 192.168.10.25
- Account: Administrator

- Event ID: 4624
- Authentication following repeated failures

### Severity

### High

### Recommended Investigation Steps

1. Verify whether the login was expected.
  2. Identify the Logon Type.
  3. Review all actions performed after login.
  4. Check privilege escalation events.
  5. Look for new user creation.
  6. Review process creation events.
  7. Determine whether MFA was enabled (if applicable).
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### Event 3

### Sysmon Event ID 1

### powershell.exe executed with an encoded command

### Explanation

Sysmon Event ID 1 records process creation.

PowerShell executed with an **EncodedCommand** parameter is a common attacker technique used to:

- Hide malicious commands
- Execute malware
- Download payloads
- Bypass simple detection mechanisms

Although administrators sometimes use encoded commands legitimately, they are uncommon in normal daily operations.

### Suspicious Indicators

- Encoded PowerShell command
- Obfuscated execution
- Execution shortly after Administrator login
- Possible malware execution

### Indicators of Compromise (IoCs)

- Process: powershell.exe

- EncodedCommand parameter
- Parent process (to be investigated)
- User context: Administrator

## **Severity**

## **High**

## **Recommended Investigation Steps**

1. Extract and decode the Base64-encoded PowerShell command.
2. Identify the parent process.
3. Review child processes spawned by PowerShell.
4. Check for persistence mechanisms.
5. Search for downloaded files.
6. Review PowerShell operational logs.
7. Compare execution against known administrative scripts.

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## **Event 4**

### **Sysmon Event ID 3**

#### **powershell.exe established an outbound HTTPS connection**

**Destination IP:** 203.0.113.50

**Port:** 443

#### **Explanation**

Sysmon Event ID **3** records network connections.

PowerShell initiating an outbound HTTPS connection is suspicious because attackers commonly use encrypted channels for:

- Command-and-Control (C2)
- Malware downloads
- Data exfiltration
- Beaconing

The destination IP should be investigated immediately.

#### **Suspicious Indicators**

- PowerShell initiating network communication
- External destination
- HTTPS encrypted traffic
- Connection immediately after encoded PowerShell execution

## Indicators of Compromise (IoCs)

- Destination IP: 203.0.113.50
- Port: 443
- Process: powershell.exe
- Outbound encrypted connection

## Severity

### Critical

## Recommended Investigation Steps

1. Perform reputation checks on the destination IP.
2. Review firewall and proxy logs for additional connections.
3. Determine whether data was transferred.
4. Capture network traffic if available.
5. Check for recurring beaconing activity.
6. Search for the same destination IP across the environment.
7. Isolate the affected host if malicious activity is confirmed.

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## Overall Assessment

### Timeline

| Time              | Event                                | Assessment                                      |
|-------------------|--------------------------------------|-------------------------------------------------|
| T0                | Multiple failed Administrator logins | Possible brute-force/password guessing          |
| T0 + 5 min        | Successful Administrator login       | Potential account compromise                    |
| Immediately after | Encoded PowerShell execution         | Likely malicious code execution                 |
| Shortly after     | PowerShell outbound HTTPS connection | Possible Command-and-Control (C2) communication |

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## Correlated Findings

The events form a coherent attack chain:

1. An attacker repeatedly attempted to authenticate to the **Administrator** account from 192.168.10.25.
2. A successful login from the same IP shortly afterward raises the likelihood of credential compromise.

3. The authenticated session launched **PowerShell** with an encoded command, indicating possible execution of obfuscated or malicious code.
4. The PowerShell process then established an encrypted outbound connection to 203.0.113.50 over HTTPS, which is consistent with potential **Command-and-Control (C2)** activity or payload retrieval.

### **Overall Severity**

#### **Critical**

The combination of:

- Repeated failed logins,
- A subsequent successful privileged login,
- Encoded PowerShell execution, and
- External HTTPS communication

strongly suggests a likely compromise that requires immediate investigation.

#### **Immediate Response Actions**

1. Isolate the affected endpoint from the network.
2. Disable or reset the **Administrator** account credentials if compromise is suspected.
3. Block communication with 203.0.113.50 at the firewall or endpoint protection platform.
4. Acquire volatile evidence (running processes, active network connections, memory) if possible.
5. Collect Windows Security, Sysmon, PowerShell, and EDR logs for forensic analysis.
6. Hunt for similar PowerShell executions, failed/successful login patterns, and connections to the same destination IP across other systems.
7. Escalate the incident to the Tier-2 SOC or Incident Response team for containment and deeper forensic investigation.